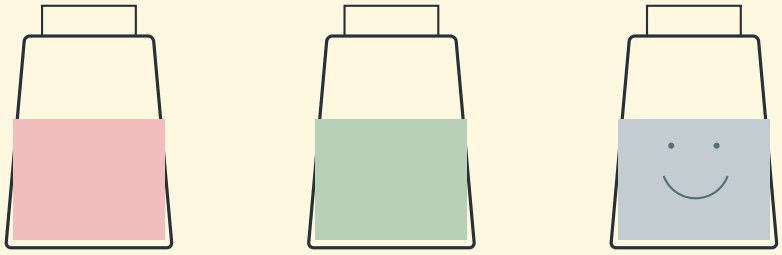
ACIDIC, BASIC & NEUTRAL

Student-Friendly Notes

Class 7 Science — Exploring Substances



ACID
blue litmus → red

BASE
red litmus → blue

NEUTRAL
no litmus change

Based on the uploaded Grade 7 *Curiosity* textbook, Chapter 2.
Includes **all 12 end-of-chapter questions**, the **Dive Deeper** question, activity answers, indicator tables, diagrams, memory tricks, and common mistakes.

Golden Rule: Never taste an unknown substance to decide whether it is acidic or basic. Use an indicator.

Editable Overleaf project — diagrams are made with TikZ, so they stay sharp and editable.

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1. All Textbook Exercise Questions — Solved

The following answers cover every numbered question under *Let Us Enhance Our Learning* and the final *Dive Deeper* question.

Q1. Reversing the red-litmus-to-blue change

If a solution turns red litmus blue, it is basic. To reverse the change, we need an acid in excess. Among the options, **vinegar** is acidic.
Answer: (iii) Vinegar.

Q2. Identify A, B and C from indicator results

A: red litmus turns blue ⇒ basic.
B: turmeric turns red ⇒ basic.
C: red rose extract turns green ⇒ basic.
Answer: (iv) Basic, basic, and basic.

Q3. Nature of the solutions in Figs. 2.13–2.15

Use the red-rose indicator rule: **green = basic**, **red = acidic**, and **no indicator colour change = neutral**.
From the colours shown in the textbook figures:

- **Fig. 2.13: Basic**
- **Fig. 2.14: Neutral**
- **Fig. 2.15: Acidic**

Q4. Red litmus unchanged, blue litmus turns red, turmeric unchanged

The liquid is **acidic**. The strongest evidence is that **blue litmus turns red**. Red litmus remains red in an acid, and turmeric also does not change colour in an acidic solution.

Q5. Manya is blindfolded — which indicator?

Manya should use an **olfactory indicator**, such as **onion**, because it can be tested by smell instead of colour. In the usual onion-indicator test, the onion smell is retained in an acidic medium but is not detected in a basic medium. This lets her distinguish the two without seeing a colour change.

Q6. Secret-message combinations

Several indicator/reagent combinations can reveal writing. Examples are:

Writing material	Spray / indicator	Visible result
Soap solution (basic)	Turmeric solution	Writing turns red on yellow background
Baking soda solution (basic)	Turmeric solution	Writing turns red
Lime water (basic)	Red rose extract	Writing shows a green shade
Vinegar (acidic)	Blue litmus solution	Acidic writing changes indicator from blue to red
Lemon juice (acidic)	Red rose extract	Acidic writing gives a red shade relative to the indicator colour

The chapter itself gives one clear possibility: **write with soap solution and use turmeric as the revealing indicator**.

Q7. Grape juice + red rose extract, then baking soda

The red tint indicates that the grape-juice mixture is acidic. Baking soda is basic, so it will **neutralise the acid**. As more baking soda is added, the acidic red response will weaken; if enough/excess base is present, the red-rose indicator will show its **basic green shade**.

Q8. Reveal a secret message written with orange juice

Orange juice is acidic. Use a **blue litmus solution** as the revealing indicator: where it contacts the acidic orange-juice writing, the blue litmus turns **red**, making the message visible.
Why not turmeric? Turmeric does not distinguish an acid from a neutral substance.

Q9. How can a natural indicator be prepared?

Example: red rose indicator

1. Collect fallen red-rose petals and wash them.
2. Crush the petals with a mortar and pestle.
3. Put them in a glass tumbler and add enough hot water to immerse them.
4. Cover and wait about 5–10 minutes for the water to become coloured.
5. Filter the mixture. The coloured filtrate is the natural indicator.

It turns towards a red shade in acidic solutions and towards green in basic solutions.

Q10. Vinegar, baking soda solution and sugar solution — turmeric only?

No, all three cannot be identified using only turmeric paper.
Baking soda solution is basic, so turmeric turns red and identifies it. But both vinegar (acidic) and sugar solution (neutral) leave turmeric yellow. Therefore turmeric alone cannot tell which of the remaining two is vinegar and which is sugar solution.

Q11. Red rose extract turns X green; then add excess amla juice

If red rose extract turns liquid X green, **X is basic**. Amla juice is acidic. Adding it first neutralises the basic liquid. When **amla juice is in excess**, the mixture becomes acidic, so the red-rose indicator shows its **acidic red shade**.

Q12. Complete the soil flowchart

A suitable completion is:

- The soil can be **acidic** in nature.
- The soil can be **basic** in nature.
- Use **red and blue litmus paper (or another acid-base indicator)** to test the soil extract.
- Acidic soil can be treated with **lime**, which is basic.
- Basic soil can be treated with **organic matter such as manure or composted leaves**, which releases acids and helps neutralise it.

Dive Deeper. Why does vinegar bubble with eggshell/marble, but soap solution does not?

Eggshell and marble contain mainly **calcium carbonate**. Vinegar is acidic, so it reacts with calcium carbonate and produces **carbon dioxide gas**; the escaping gas appears as bubbles. Soap solution is basic and does not produce the same acid-carbonate reaction, so similar bubbling is not seen.
Exam phrase: *The bubbles are carbon dioxide gas released when the acid in vinegar reacts with calcium carbonate.*

2. Acidic, Basic and Neutral Substances

Substances around us can be grouped as **acidic**, **basic**, or **neutral**. We usually identify their nature using **indicators** rather than by tasting them.

ACIDIC
blue litmus → red

BASIC
red litmus → blue

NEUTRAL
no litmus colour change

Key Idea

An **acid-base indicator** is a substance that shows different colours in acidic and basic solutions. Litmus is a natural indicator obtained from lichens.

What the textbook samples show

Sample	Blue litmus	Red litmus	Nature
Lemon juice	Turns red	No change	Acidic
Soap solution	No change	Turns blue	Basic
Amla juice	Turns red	No change	Acidic
Tamarind water	Turns red	No change	Acidic
Vinegar	Turns red	No change	Acidic
Baking soda solution	No change	Turns blue	Basic
Lime water	No change	Turns blue	Basic
Tap water	No change	No change	Neutral
Washing powder solution	No change	Turns blue	Basic
Sugar solution	No change	No change	Neutral
Salt solution	No change	No change	Neutral

Memory Trick

A-B shortcut: **A**cid turns **B**lue litmus red. A **B**ase does the opposite: red litmus becomes blue.

Common Mistake

Do not say “all sour things are safe acids” or “all bitter things are bases.” The textbook says edible acidic substances often taste sour and bases generally taste bitter/slippery, but unknown substances must **never** be tasted.

Common acids in foods

The textbook asks students to name the common acids in familiar substances. The standard school-level answers are:

Substance	Common acid	Easy clue
Lemon	Citric acid	citrus → citric
Curd	Lactic acid	milk/curd → lactic
Tamarind	Tartaric acid	tamarind → tartaric
Vinegar	Acetic acid	vinegar smell/taste → acetic

3. Indicators: Colour Detectives

Litmus

Blue litmus
Acid makes it red

Red litmus
Base makes it blue

Neutral solution
Neither strip changes

Red rose extract

The textbook prepares red rose extract by crushing washed petals, soaking them in hot water, waiting 5–10 minutes, and filtering the coloured liquid.

Acidic solution
red shade

Neutral solution
no indicator change

Basic solution
green shade

Turmeric paper

Turmeric paper is made by coating filter paper with turmeric paste, drying it, and cutting it into strips.

Acidic or neutral
yellow stays yellow

Basic
yellow turns red

Key Idea

Turmeric can identify a **base**, but it cannot distinguish an **acid** from a **neutral** solution because both leave turmeric yellow.

Other natural and olfactory indicators

The chapter also mentions red hibiscus, purple cabbage, beetroot, jamun and turmeric as natural indicator sources. **Olfactory indicators** are substances whose odour changes in acidic or basic media. The activity uses onion-soaked cloth/paper strips.

Blindfold-friendly clue

A student who cannot rely on colour can use an olfactory indicator such as onion. In the usual school demonstration, onion smell is retained in an acidic medium but is not detected in a basic medium.

4. Textbook Activities — What Should You Observe?

Activity 2.1: Litmus sorting

Group A (acidic): lemon juice, amla juice, tamarind water, vinegar.
Group B (basic): soap solution, baking soda solution, lime water, washing powder solution.
Group C (neutral): tap water, sugar solution, salt solution.

Activities 2.3–2.4: Red rose indicator

Activity / Observation

Lemon juice and other acidic samples turn the red rose extract towards a **red shade**. Soap solution and other basic samples turn it towards a **green shade**. Neutral samples do not show the acid/base colour change.

Activity 2.5: Turmeric

Basic samples change turmeric paper from yellow to **red**. Acidic and neutral samples do not change its yellow colour. Therefore turmeric is useful for detecting bases but not for separating acids from neutral substances.

Secret-message idea from the chapter

If a message is written with a basic solution such as soap solution and the sheet is treated with turmeric, the writing appears red while the rest remains yellow.



5. Neutralisation

When an acid and a base are mixed in suitable quantities, they neutralise each other.

$$\textbf{Acid} + \textbf{Base} \longrightarrow \textbf{Salt} + \textbf{Water} + \textbf{Heat}$$

Activity / Observation

In the textbook experiment, diluted lemon juice first makes blue litmus solution red. Lime water is then added. As enough base is added, the acidic effect is neutralised; with further base the indicator shows a basic condition. This demonstrates that an acid and a base can neutralise one another.

Memory Trick

N = Neither: at neutralisation, the resulting solution is neither acidic nor basic (when acid and base are present in the correct amounts). Salt and water form, and heat is released.

6. Neutralisation in Daily Life

Situation	Problem	What helps?
Red ant bite	Ant injects acidic formic acid	Moist baking soda, a base, neutralises the acidic effect
Acidic soil	Excessive fertiliser can make soil acidic	Lime, a base, helps neutralise it
Basic soil	Soil is too basic for healthy growth	Organic matter such as manure/composted leaves releases acids and helps neutralise it
Acidic factory waste	Acidic waste may harm life in a lake	Neutralise with a suitable basic substance before release

Everyday Science

A poor-looking plant does **not automatically prove** that the soil is acidic or basic. The chapter also warns that poor plant health can be due to nutrient deficiency even when soil is neutral.

7. One-Page Revision Map

Exploring Substances: Acidic, Basic, and Neutral

ACIDIC

- Blue litmus → red
- Red rose → red shade
- Turmeric: no change
- Examples: lemon, amla, tamarind, vinegar

BASIC

- Red litmus → blue
- Red rose → green shade
- Turmeric yellow → red
- Examples: soap, baking soda, lime water

NEUTRAL

- No change in red/blue litmus
- Red rose: no acid/base change
- Turmeric: no change
- Examples: textbook lists tap water, sugar solution, salt solution

NEUTRALISATION

Acid + Base → Salt + Water + Heat

- ant bite → baking soda
- acidic soil → lime
- basic soil → organic matter
- acidic waste → neutralise before release

Memory Trick

Indicator triangle: Litmus works for acid/base, red rose gives different colours for acid/base, and turmeric is especially good for spotting a base.

8. In-Chapter Questions — Quick Answers

Can hydrangea colour be altered by changing soil nature?

The chapter notes that hydrangea can produce blue flowers in acidic soil and pink/red flowers in basic soil. So, in principle, changing the soil's acid/base nature can change flower colour.

What solution can be used to write on turmeric-coated paper?

Use a **basic solution**, such as soap solution or baking soda solution, because turmeric turns red in a basic medium.

Why did moist baking soda relieve an ant bite?

The ant injects acidic formic acid. Baking soda is basic, so it neutralises the acidic effect.

What should be done with acidic industrial waste before it reaches a lake?

Neutralise it with a suitable basic substance before discharge, as described in the chapter.

How did the science-fair secret message appear?

One textbook possibility is that the writing was made with a **basic soap solution** and the paper was sprayed/treated with **turmeric indicator**; the basic writing changes turmeric's colour and becomes visible.

9. Common Mistakes to Avoid

Common Mistake

- **Wrong:** Acid turns red litmus blue. **Correct:** acid turns **blue** litmus **red**.
- **Wrong:** Turmeric identifies acids. **Correct:** turmeric is especially useful for identifying **bases**; acid and neutral both show no turmeric colour change.
- **Wrong:** Neutralisation makes only water. **Correct:** it forms **salt + water** and releases **heat**.
- **Wrong:** A plant looking weak proves soil is acidic. **Correct:** soil could also be basic, neutral, or nutrient-deficient.
- **Wrong:** Taste an unknown liquid to check if it is sour/bitter. **Correct:** never taste unknown substances; use indicators.

10. Rapid Revision Test

1. Which litmus paper changes colour in an acid? *Blue litmus; it turns red.*
2. Which litmus paper changes colour in a base? *Red litmus; it turns blue.*
3. What colour does red-rose indicator show in a base? *Green shade.*
4. What happens to turmeric in a base? *Yellow turns red.*
5. Can turmeric distinguish vinegar from sugar solution? *No.*
6. Write the neutralisation word equation. *Acid + Base → Salt + Water + Heat.*
7. What can neutralise acidic soil? *Lime.*
8. What can help neutralise basic soil? *Organic matter such as manure/composted leaves.*

9. Why does vinegar bubble on marble/eggshell? *Acid reacts with calcium carbonate and releases carbon dioxide.*

If you remember the indicator colour changes + the neutralisation equation, most questions in this chapter become reasoning questions rather than memorisation questions.